Technical Assessment

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Preliminary setup

In the `stevedata` package in R, we load in the wage data from the General Social Survey (1974-2018), illustrating wage gap by gender. (See here for more detailed documentation.) After we load in the data, we display some summary statistics to begin with.

```
# some helpful libraries
library(stevedata)
library(dplyr)
library(ggplot2)
library(reshape2)
```

```
# load in the data & display summary statistics
data(gss_wages)
summary(gss_wages)
```

```
      year         realrinc            age             occ10     
 Min.   :1974   Min.   :   227   Min.   :18.00   Min.   :  10  
 1st Qu.:1985   1st Qu.:  8156   1st Qu.:32.00   1st Qu.:2710  
 Median :1996   Median : 16563   Median :44.00   Median :4720  
 Mean   :1996   Mean   : 22326   Mean   :46.18   Mean   :4696  
 3rd Qu.:2006   3rd Qu.: 27171   3rd Qu.:59.00   3rd Qu.:6230  
 Max.   :2018   Max.   :480144   Max.   :89.00   Max.   :9997  
                NA's   :23810    NA's   :219     NA's   :3561  
  occrecode            prestg10         childs         wrkstat         
 Length:61697       Min.   :16.00   Min.   :0.000   Length:61697      
 Class :character   1st Qu.:33.00   1st Qu.:0.000   Class :character  
 Mode  :character   Median :42.00   Median :2.000   Mode  :character  
                    Mean   :43.06   Mean   :1.923                     
                    3rd Qu.:50.00   3rd Qu.:3.000                     
                    Max.   :80.00   Max.   :8.000                     
```

```
                    NA's   :4186    NA's   :189     
    gender            educcat           maritalcat       
 Length:61697       Length:61697       Length:61697      
 Class :character   Class :character   Class :character  
 Mode  :character   Mode  :character   Mode  :character  
```

From these initial summary statistics, we see that there are a lot of missing values, so we subset our data to only include complete observations, which make up 60.27% of the original dataset.

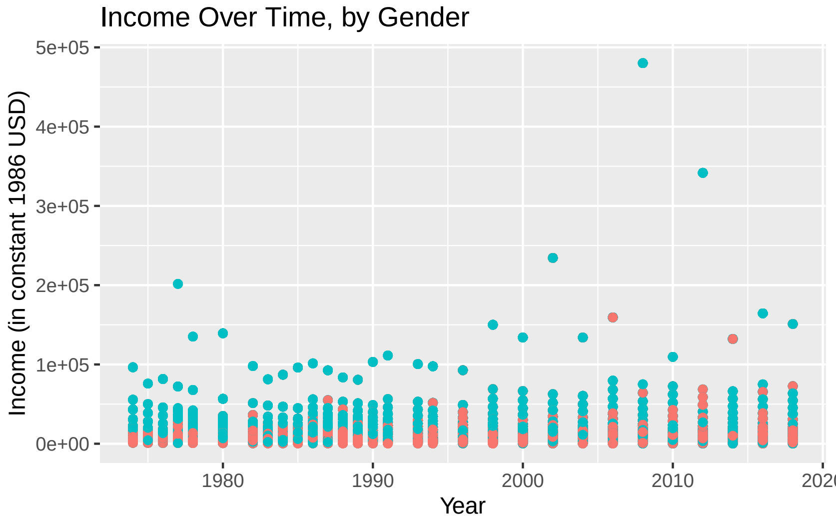
```
# filter to only include complete observations
gss_wages_complete <- gss_wages[complete.cases(gss_wages),]
paste0("Percentage complete observations: ",
       round(100*(nrow(gss_wages_complete))/nrow(gss_wages), 2), "%")
```

```
[1] "Percentage complete observations: 60.27%"
```

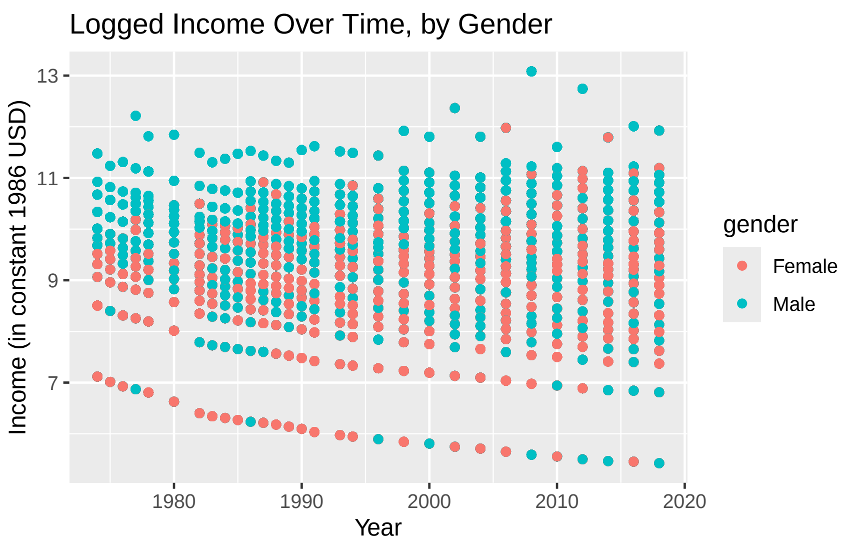
Fact 1: Females have lower income than males.

From the data documentation, we see that it aims to illuminate the wage gap between gender. We start investigating this suggestion by graphing income levels by gender over time:

```
# plot of year v.s. income
ggplot(gss_wages_complete) +
  geom_point(aes(x = year, y = realrinc, color = gender)) +
  labs(x = "Year", y = "Income (in constant 1986 USD)",
       title = "Income Over Time, by Gender") +
  theme(legend.position = "none")
```




```
# plot of year v.s. logged income
ggplot(gss_wages_complete) +
  geom_point(aes(x = year, y = log(realrinc), color = gender)) +
  labs(x = "Year", y = "Income (in constant 1986 USD)",
       title = "Logged Income Over Time, by Gender")
```



The initial plot seems to suggest that males (blue) have higher income than females (red), demonstrated by all the top-most dots being the color blue. To take care of outliers and get a better visual of the pattern, we can also plot the logged income values; this graph shows with a little more clarity the trend again of males making more than females.

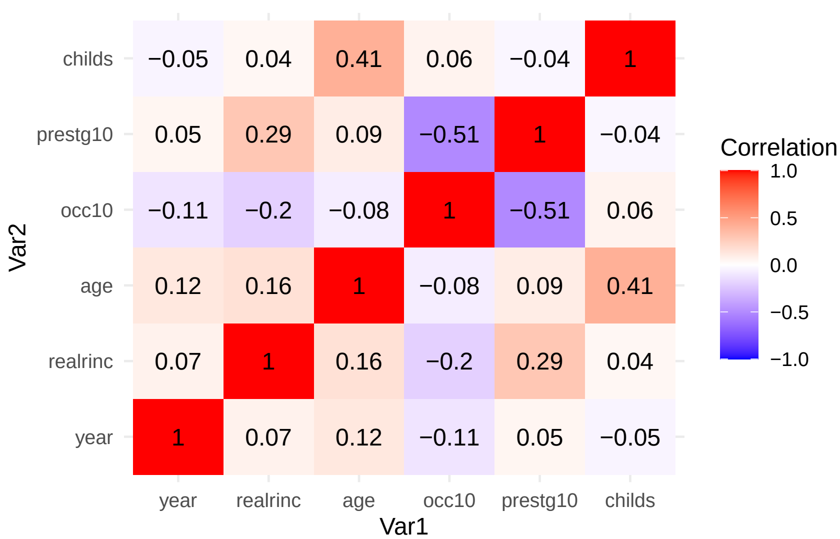
This is important because gender pay equality is something that is supposed to be getting better over time, but this graph suggests that there is still a marked difference.

Fact 2: The number of children has a very small correlation with base income.

From here, one might wonder if any other variables have correlations with each other, or with income in particular. To check this out, we plot a heatmap of correlation values:

```
# correlation matrix
corrs <- gss_wages_complete |>
  select(c(year, realrinc, age, occ10, prestg10, childs)) |>
  cor()

# from matrix, plot heatmap
corrs |>
  melt() |>
  ggplot(aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  geom_text(aes(label = round(value, 2))) +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
                       limit = c(-1, 1), name = "Correlation") +
  theme_minimal()
```



While one might expect that parents suffer in their careers and therefore income levels to take care of children, we see in particular that `childs` only has a correlation of 0.04 with `realrinc`. Because of this counterintuitive idea, one might hypothesize further that perhaps only *young* children negatively impact income, a distinction that this plot is unable to capture. Additional investigation would be needed. The other variables do not have particularly high correlations with one another either, save for prestige score and occupation, which makes sense (people may see higher paying jobs as more prestigious.)

This is important to investigate because a drop in income due to having children may decentivize individuals trying to raise families, a very pertinent issue in the light of dropping birthrates.